MIDDLESEX SCHOOL

MATHEMATICS CURRICULUM REPORT

Math 32

Precalculus: Trigonometry

MIDDLESEX SCHOOL | MATHEMATICS

Math 32 Course at a Glance

Precalculus: Trigonometry

Trigonometric Foundations

REQUIRED

LEARNING OBJECTIVES

M32-FND-001	I can use the Pythagorean theorem to determine an unknown side of a right triangle.
M32-FND-002	I can determine exact side lengths in 30-60-90 and 45-45-90 triangles.
M32-FND-003	I can convert angle measures between degrees and radians.
M32-FND-004	I can determine coterminal and reference angles for a given angle.
M32-FND-005	I can calculate arc length from a central angle and radius.
M32-FND-006	I can determine exact coordinates on the unit circle.
M32-FND-007	I can evaluate all six trigonometric functions exactly for common angles.
M32-FND-008	I can solve right triangles using trigonometric ratios.
M32-FND-009	I can apply right-triangle trigonometry to contextual problems.
M32-FND-010	I can evaluate trigonometric functions from a point on a circle or terminal side.
M32-FND-011	I can solve a basic trigonometric equation for all angles in a specified interval.

Periodic Functions and Modeling

REQUIRED

LEARNING OBJECTIVES

M32-GRF-001	I can graph basic sine and cosine functions using key features.
M32-GRF-002	I can identify amplitude, midline, period, and phase shift from an equation or graph.
M32-GRF-003	I can graph transformed sine and cosine functions.
M32-GRF-004	I can write sine and cosine equations that represent a periodic graph.
M32-GRF-005	I can construct a sinusoidal model from contextual information or data.
M32-GRF-006	I can interpret the parameters and outputs of a sinusoidal model in context.
M32-GRF-007	I can graph tangent, secant, and cosecant functions with their asymptotes.

Inverse Functions and Trigonometric Equations

REQUIRED

LEARNING OBJECTIVES

M32-INV-001	I can evaluate inverse sine, inverse cosine, and inverse tangent expressions.
M32-INV-002	I can explain how restricted domains make inverse trigonometric functions possible.
M32-INV-003	I can solve trigonometric equations exactly or approximately on a specified interval.
M32-INV-004	I can express all solutions of a trigonometric equation in general form.
M32-INV-005	I can evaluate compositions of trigonometric and inverse trigonometric functions.

Non-Right Triangles

REQUIRED

LEARNING OBJECTIVES

M32-TRI-002	I can solve a non-right triangle using the Law of Sines.
M32-TRI-003	I can determine the number of possible triangles in the ambiguous SSA case.
M32-TRI-004	I can solve a non-right triangle using the Law of Cosines.
M32-TRI-005	I can select and apply an appropriate method to solve a triangle in context.

Math 32 Course at a Glance

Precalculus: Trigonometry

Trigonometric Identities

REQUIRED

LEARNING OBJECTIVES

M32-IDN-001	I can rewrite trigonometric expressions using reciprocal and quotient identities, including `tan(theta) = sin(theta)/cos(theta)`.
M32-IDN-002	I can rewrite trigonometric expressions using the Pythagorean identities.
M32-IDN-003	I can use even/odd and cofunction identities to rewrite trigonometric expressions.
M32-IDN-004	I can verify a trigonometric identity using the required identity families.

Additional Trigonometric Identities

EXTENSIO
N

LEARNING OBJECTIVES

M32-IDX-001	I can apply double-angle identities.
M32-IDX-002	I can apply half-angle identities.
M32-IDX-003	I can apply angle sum and difference identities.
M32-IDX-004	I can determine exact trigonometric values using extension identities.

Triangle Area from SAS

EXTENSIO
N

LEARNING OBJECTIVES

M32-TRX-001	I can calculate the area of a triangle from two sides and the included angle.

Polar Coordinates

REQUIRED

LEARNING OBJECTIVES

M32-POL-001	I can plot and identify points in polar coordinates.
M32-POL-002	I can represent a polar point using equivalent coordinate pairs.
M32-POL-003	I can convert points between polar and rectangular coordinates.

Math 32 Course at a Glance

Precalculus: Trigonometry

Polar Functions

EXTENSIO
N

LEARNING OBJECTIVES

M32-PLX-001	I can graph and interpret basic polar functions.
M32-PLX-002	I can convert equations between polar and rectangular forms.

Vectors

EXTENSIO
N

LEARNING OBJECTIVES

M32-VEC-001	I can represent a vector geometrically and in component form.
M32-VEC-002	I can add, subtract, and multiply vectors by scalars.
M32-VEC-003	I can determine vector magnitude, direction, and unit vector.
M32-VEC-004	I can resolve a vector into horizontal and vertical components.
M32-VEC-005	I can perform basic operations with vectors in three dimensions.

Parametric Equations

EXTENSIO
N

LEARNING OBJECTIVES

M32-PAR-001	I can graph a plane curve defined by parametric equations with its orientation.
M32-PAR-002	I can eliminate a parameter to obtain a rectangular equation.
M32-PAR-003	I can create or interpret parametric equations for motion in the plane.

Math 32 In-Depth Curriculum

Precalculus: Trigonometry

SKILL PROGRESSION

I Introduce · R Reinforce · D Deepen · A Apply

Trigonometric Foundations

REQUIRED

LEARNING OBJECTIVES WITH SUPPORTING SKILLS

M32-FND-001 — I can use the Pythagorean theorem to determine an unknown side of a right triangle.

SUPPORTING SKILLS	
A	Extract perfect-power factors from a radical. (first introduced in Math 21) SK-RAD-FACTOR-PERFECT · Apply
R	Apply the Pythagorean theorem or its converse. (first introduced in Math 22) SK-TRI-PYTHAGOREAN · Reinforce

M32-FND-002 — I can determine exact side lengths in 30-60-90 and 45-45-90 triangles.

SUPPORTING SKILLS	
A	Distinguish exact values from measured or rounded approximations. (first introduced in Math 22) SK-GEO-EXACT-APPROX · Apply
A	Extract perfect-power factors from a radical. (first introduced in Math 21) SK-RAD-FACTOR-PERFECT · Apply
R	Recall special-right-triangle ratios. (first introduced in Math 22) SK-TRI-SPECIAL-RATIOS · Reinforce

M32-FND-003 — I can convert angle measures between degrees and radians.

SUPPORTING SKILLS	
A	Add, subtract, multiply, and divide fractions. (inherited from middle school) SK-NUM-FRACTION-OPERATE · Apply
I	Convert between degree and radian measure. SK-TRIG-ANGLE-CONVERT · Introduce

M32-FND-004 — I can determine coterminal and reference angles for a given angle.

SUPPORTING SKILLS	
I	Generate and recognize coterminal angles. SK-TRIG-ANGLE-COTERMINAL · Introduce
I	Determine a reference angle and its quadrant. SK-TRIG-ANGLE-REFERENCE · Introduce
I	Determine the sign of a trigonometric function by quadrant. SK-TRIG-SIGN-QUADRANT · Introduce

M32-FND-005 — I can calculate arc length from a central angle and radius.

SUPPORTING SKILLS	
D	Calculate arc length from radius and central angle. (first introduced in Math 22) SK-CIR-ARC-LENGTH · Deepen
A	Select and report appropriate linear, square, or angular units. (first introduced in Math 22) SK-GEO-UNITS · Apply
A	Convert between degree and radian measure. (first introduced in Math 32) SK-TRIG-ANGLE-CONVERT · Apply

M32-FND-006 — I can determine exact coordinates on the unit circle.

SUPPORTING SKILLS	
A	Recall special-right-triangle ratios. (first introduced in Math 22) SK-TRI-SPECIAL-RATIOS · Apply
A	Determine a reference angle and its quadrant. (first introduced in Math 32) SK-TRIG-ANGLE-REFERENCE · Apply
I	Determine unit-circle coordinates for common angles. SK-TRIG-UNIT-CIRCLE-COORD · Introduce

Math 32 In-Depth Curriculum

Precalculus: Trigonometry

SKILL PROGRESSION

I Introduce · R Reinforce · D Deepen · A Apply

M32-FND-007 — I can evaluate all six trigonometric functions exactly for common angles.

SUPPORTING SKILLS

I	Express common trigonometric values exactly. SK-TRIG-EXACT-VALUE · Introduce
I	Relate sine/cosecant, cosine/secant, and tangent/cotangent. SK-TRIG-RECIPROCAL · Introduce
A	Determine the sign of a trigonometric function by quadrant. (first introduced in Math 32) SK-TRIG-SIGN-QUADRANT · Apply
A	Determine unit-circle coordinates for common angles. (first introduced in Math 32) SK-TRIG-UNIT-CIRCLE-COORD · Apply

M32-FND-008 — I can solve right triangles using trigonometric ratios.

SUPPORTING SKILLS

D	Match opposite, adjacent, and hypotenuse sides to a trigonometric ratio. (first introduced in Math 22) SK-TRI-TRIG-RATIO · Deepen
D	Use an inverse trigonometric function to determine a missing angle. (first introduced in Math 22) SK-TRI-TRIG-SOLVE-ANGLE · Deepen
D	Use a trigonometric ratio to determine a missing side. (first introduced in Math 22) SK-TRI-TRIG-SOLVE-SIDE · Deepen
R	Label sides relative to a selected angle. (first introduced in Math 22) SK-TRIG-TRIANGLE-LABEL · Reinforce

M32-FND-009 — I can apply right-triangle trigonometry to contextual problems.

SUPPORTING SKILLS

A	Restrict a model to meaningful contextual inputs. (first introduced in Math 12) SK-MODEL-VALID-DOMAIN · Apply
A	Attach and interpret units for rates, inputs, and outputs. (first introduced in Math 12) SK-REP-UNITS · Apply
D	Match opposite, adjacent, and hypotenuse sides to a trigonometric ratio. (first introduced in Math 22) SK-TRI-TRIG-RATIO · Deepen
A	Use an inverse trigonometric function to determine a missing angle. (first introduced in Math 22) SK-TRI-TRIG-SOLVE-ANGLE · Apply
A	Use a trigonometric ratio to determine a missing side. (first introduced in Math 22) SK-TRI-TRIG-SOLVE-SIDE · Apply

M32-FND-010 — I can evaluate trigonometric functions from a point on a circle or terminal side.

SUPPORTING SKILLS

A	Apply the Pythagorean theorem or its converse. (first introduced in Math 22) SK-TRI-PYTHAGOREAN · Apply
I	Use coordinates and radius to evaluate trigonometric functions. SK-TRIG-CIRCLE-POINT · Introduce
A	Relate sine/cosecant, cosine/secant, and tangent/cotangent. (first introduced in Math 32) SK-TRIG-RECIPROCAL · Apply
A	Determine the sign of a trigonometric function by quadrant. (first introduced in Math 32) SK-TRIG-SIGN-QUADRANT · Apply

Math 32 In-Depth Curriculum

Precalculus: Trigonometry

SKILL PROGRESSION

I Introduce · R Reinforce · D Deepen · A Apply

M32-FND-011 **I can solve a basic trigonometric equation for all angles in a specified interval.**

SUPPORTING SKILLS

A	Translate between inequality and interval notation. (first introduced in Math 12) SK-NOT-INTERVAL · Apply
A	Determine a reference angle and its quadrant. (first introduced in Math 32) SK-TRIG-ANGLE-REFERENCE · Apply
I	Use reference angles and periodicity to find every solution in an interval. SK-TRIG-EQUATION-INTERVAL · Introduce
I	Isolate a trigonometric expression before solving. SK-TRIG-EQUATION-ISOLATE · Introduce
A	Determine the sign of a trigonometric function by quadrant. (first introduced in Math 32) SK-TRIG-SIGN-QUADRANT · Apply

Math 32 In-Depth Curriculum

Precalculus: Trigonometry

SKILL PROGRESSION

I Introduce · R Reinforce · D Deepen · A Apply

Periodic Functions and Modeling

REQUIRED

LEARNING OBJECTIVES WITH SUPPORTING SKILLS

M32-GRF-001 I can graph basic sine and cosine functions using key features.

SUPPORTING SKILLS

D	Use tabular patterns to predict graph shape and behavior. (first introduced in Math 21) SK-REP-TABLE-GRAPH · Deepen
I	Locate quarter-period key points of a periodic function. SK-TRIG-GRAPH-KEYPOINTS · Introduce
A	Determine unit-circle coordinates for common angles. (first introduced in Math 32) SK-TRIG-UNIT-CIRCLE-COORD · Apply

M32-GRF-002 I can identify amplitude, midline, period, and phase shift from an equation or graph.

SUPPORTING SKILLS

I	Determine amplitude and vertical reflection. SK-TRIG-GRAPH-AMPLITUDE · Introduce
I	Determine a sinusoid's midline and vertical shift. SK-TRIG-GRAPH-MIDLINE · Introduce
I	Determine period from a graph or equation. SK-TRIG-GRAPH-PERIOD · Introduce
I	Determine horizontal shift from a graph or equation. SK-TRIG-GRAPH-PHASE · Introduce

M32-GRF-003 I can graph transformed sine and cosine functions.

SUPPORTING SKILLS

A	Interpret transformations applied to function inputs. (first introduced in Math 31) SK-FUNC-TRANSFORM-HORIZONTAL · Apply
D	Distinguish stretches from compressions and horizontal effects from vertical effects. (first introduced in Math 31) SK-FUNC-TRANSFORM-SCALE · Deepen
A	Interpret transformations applied to function outputs. (first introduced in Math 31) SK-FUNC-TRANSFORM-VERTICAL · Apply
D	Locate quarter-period key points of a periodic function. (first introduced in Math 32) SK-TRIG-GRAPH-KEYPOINTS · Deepen

M32-GRF-004 I can write sine and cosine equations that represent a periodic graph.

SUPPORTING SKILLS

D	Connect parameters in a formula to features of its graph. (first introduced in Math 21) SK-REP-FORMULA-GRAPH · Deepen
D	Determine amplitude and vertical reflection. (first introduced in Math 32) SK-TRIG-GRAPH-AMPLITUDE · Deepen
D	Determine a sinusoid's midline and vertical shift. (first introduced in Math 32) SK-TRIG-GRAPH-MIDLINE · Deepen
D	Determine period from a graph or equation. (first introduced in Math 32) SK-TRIG-GRAPH-PERIOD · Deepen
D	Determine horizontal shift from a graph or equation. (first introduced in Math 32) SK-TRIG-GRAPH-PHASE · Deepen

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Precalculus: Trigonometry

SKILL PROGRESSION

I Introduce · R Reinforce · D Deepen · A Apply

M32-GRF-005 — I can construct a sinusoidal model from contextual information or data.

SUPPORTING SKILLS

A	Restrict a model to meaningful contextual inputs. (first introduced in Math 12) SK-MODEL-VALID-DOMAIN · Apply
A	Determine amplitude and vertical reflection. (first introduced in Math 32) SK-TRIG-GRAPH-AMPLITUDE · Apply
A	Determine a sinusoid's midline and vertical shift. (first introduced in Math 32) SK-TRIG-GRAPH-MIDLINE · Apply
A	Determine period from a graph or equation. (first introduced in Math 32) SK-TRIG-GRAPH-PERIOD · Apply
I	Translate contextual maximum, minimum, cycle, and starting position into model parameters. SK-TRIG-MODEL-PARAMETERS · Introduce

M32-GRF-006 — I can interpret the parameters and outputs of a sinusoidal model in context.

SUPPORTING SKILLS

A	Identify input and output values in a formula, graph, table, or context. (first introduced in Math 12) SK-FUNC-INPUT-OUTPUT · Apply
D	Explain a model parameter in the language of its context. (first introduced in Math 21) SK-MODEL-PARAMETER-INTERPRET · Deepen
A	Attach and interpret units for rates, inputs, and outputs. (first introduced in Math 12) SK-REP-UNITS · Apply
D	Translate contextual maximum, minimum, cycle, and starting position into model parameters. (first introduced in Math 32) SK-TRIG-MODEL-PARAMETERS · Deepen

M32-GRF-007 — I can graph tangent, secant, and cosecant functions with their asymptotes.

SUPPORTING SKILLS

A	Interpret transformations applied to function inputs. (first introduced in Math 31) SK-FUNC-TRANSFORM-HORIZONTAL · Apply
I	Locate vertical asymptotes of tangent, secant, and cosecant. SK-TRIG-GRAPH-ASYMPTOTE · Introduce
A	Locate quarter-period key points of a periodic function. (first introduced in Math 32) SK-TRIG-GRAPH-KEYPOINTS · Apply
D	Relate sine/cosecant, cosine/secant, and tangent/cotangent. (first introduced in Math 32) SK-TRIG-RECIPROCAL · Deepen

SKILL PROGRESSION

I Introduce · R Reinforce · D Deepen · A Apply

Inverse Functions and Trigonometric Equations

REQUIRED

LEARNING OBJECTIVES WITH SUPPORTING SKILLS

M32-INV-001 I can evaluate inverse sine, inverse cosine, and inverse tangent expressions.

SUPPORTING SKILLS

A	Express common trigonometric values exactly. (first introduced in Math 32) SK-TRIG-EXACT-VALUE · Apply
I	Evaluate an inverse trigonometric expression exactly or approximately. SK-TRIG-INVERSE-EVALUATE · Introduce
I	Recall the principal-value ranges of inverse trigonometric functions. SK-TRIG-INVERSE-RANGE · Introduce

M32-INV-002 I can explain how restricted domains make inverse trigonometric functions possible.

SUPPORTING SKILLS

A	Exchange domain and range between a function and its inverse. (first introduced in Math 31) SK-FUNC-INVERSE-DOMAIN-RANGE · Apply
A	Apply the horizontal line test for invertibility. (first introduced in Math 31) SK-FUNC-INVERSE-HLT · Apply
A	Test whether a function is one-to-one on a domain. (first introduced in Math 31) SK-FUNC-ONE-TO-ONE · Apply
D	Recall the principal-value ranges of inverse trigonometric functions. (first introduced in Math 32) SK-TRIG-INVERSE-RANGE · Deepen

M32-INV-003 I can solve trigonometric equations exactly or approximately on a specified interval.

SUPPORTING SKILLS

A	Check a proposed solution in the original equation. (first introduced in Math 12) SK-ALG-CHECK-SOLUTION · Apply
A	Determine a reference angle and its quadrant. (first introduced in Math 32) SK-TRIG-ANGLE-REFERENCE · Apply
D	Use reference angles and periodicity to find every solution in an interval. (first introduced in Math 32) SK-TRIG-EQUATION-INTERVAL · Deepen
D	Isolate a trigonometric expression before solving. (first introduced in Math 32) SK-TRIG-EQUATION-ISOLATE · Deepen
A	Evaluate an inverse trigonometric expression exactly or approximately. (first introduced in Math 32) SK-TRIG-INVERSE-EVALUATE · Apply

M32-INV-004 I can express all solutions of a trigonometric equation in general form.

SUPPORTING SKILLS

D	Generate and recognize coterminal angles. (first introduced in Math 32) SK-TRIG-ANGLE-COTERMINAL · Deepen
I	Express a periodic solution family in general form. SK-TRIG-EQUATION-GENERAL · Introduce
A	Use reference angles and periodicity to find every solution in an interval. (first introduced in Math 32) SK-TRIG-EQUATION-INTERVAL · Apply

Math 32 In-Depth Curriculum

Precalculus: Trigonometry

SKILL PROGRESSION

I Introduce · R Reinforce · D Deepen · A Apply

M32-INV-005	I can evaluate compositions of trigonometric and inverse trigonometric functions.
SUPPORTING SKILLS	
A	Substitute one function formula into another. (first introduced in Math 31) SK-FUNC-COMPOSE-FORMULA · Apply
A	Verify inverse functions using composition. (first introduced in Math 31) SK-FUNC-INVERSE-VERIFY · Apply
D	Evaluate an inverse trigonometric expression exactly or approximately. (first introduced in Math 32) SK-TRIG-INVERSE-EVALUATE · Deepen
A	Recall the principal-value ranges of inverse trigonometric functions. (first introduced in Math 32) SK-TRIG-INVERSE-RANGE · Apply

Math 32 In-Depth Curriculum

Precalculus: Trigonometry

SKILL PROGRESSION

I Introduce · R Reinforce · D Deepen · A Apply

Non-Right Triangles

REQUIRED

LEARNING OBJECTIVES WITH SUPPORTING SKILLS

M32-TRI-0 02 — I can solve a non-right triangle using the Law of Sines.

SUPPORTING SKILLS

A	Evaluate an inverse trigonometric expression exactly or approximately. (first introduced in Math 32) SK-TRIG-INVERSE-EVALUATE · Apply
I	Apply the Law of Sines to determine missing sides or angles. SK-TRIG-LAW-SINES · Introduce
D	Classify triangle data as SSS, SAS, ASA, AAS, or SSA. (first introduced in Math 32) SK-TRIG-TRIANGLE-DATA · Deepen

M32-TRI-0 03 — I can determine the number of possible triangles in the ambiguous SSA case.

SUPPORTING SKILLS

I	Test and interpret the possible outcomes of SSA data. SK-TRIG-AMBIGUOUS-CASE · Introduce
D	Apply the Law of Sines to determine missing sides or angles. (first introduced in Math 32) SK-TRIG-LAW-SINES · Deepen
A	Classify triangle data as SSS, SAS, ASA, AAS, or SSA. (first introduced in Math 32) SK-TRIG-TRIANGLE-DATA · Apply

M32-TRI-0 04 — I can solve a non-right triangle using the Law of Cosines.

SUPPORTING SKILLS

A	Evaluate an inverse trigonometric expression exactly or approximately. (first introduced in Math 32) SK-TRIG-INVERSE-EVALUATE · Apply
I	Apply the Law of Cosines to determine missing sides or angles. SK-TRIG-LAW-COSINES · Introduce
D	Classify triangle data as SSS, SAS, ASA, AAS, or SSA. (first introduced in Math 32) SK-TRIG-TRIANGLE-DATA · Deepen

M32-TRI-0 05 — I can select and apply an appropriate method to solve a triangle in context.

SUPPORTING SKILLS

A	Attach and interpret units for rates, inputs, and outputs. (first introduced in Math 12) SK-REP-UNITS · Apply
A	Match opposite, adjacent, and hypotenuse sides to a trigonometric ratio. (first introduced in Math 22) SK-TRI-TRIG-RATIO · Apply
A	Apply the Law of Cosines to determine missing sides or angles. (first introduced in Math 32) SK-TRIG-LAW-COSINES · Apply
A	Apply the Law of Sines to determine missing sides or angles. (first introduced in Math 32) SK-TRIG-LAW-SINES · Apply
D	Classify triangle data as SSS, SAS, ASA, AAS, or SSA. (first introduced in Math 32) SK-TRIG-TRIANGLE-DATA · Deepen
I	Select right-triangle ratios, Law of Sines, or Law of Cosines from the given data. SK-TRIG-TRIANGLE-METHOD · Introduce

SKILL PROGRESSION

I Introduce · R Reinforce · D Deepen · A Apply

Trigonometric Identities

REQUIRED

LEARNING OBJECTIVES WITH SUPPORTING SKILLS

M32-IDN-001 — **I can rewrite trigonometric expressions using reciprocal and quotient identities, including `tan(theta) = sin(theta)/cos(theta)`.**

SUPPORTING SKILLS

I	Apply reciprocal and quotient identities. SK-TRIG-IDENTITY-RECIPROCAL-QUOTIENT · Introduce
I	Substitute an identity to rewrite an expression. SK-TRIG-IDENTITY-SUBSTITUTE · Introduce

M32-IDN-002 — **I can rewrite trigonometric expressions using the Pythagorean identities.**

SUPPORTING SKILLS

I	Apply the Pythagorean trigonometric identities. SK-TRIG-IDENTITY-PYTHAGOREAN · Introduce
D	Substitute an identity to rewrite an expression. (first introduced in Math 32) SK-TRIG-IDENTITY-SUBSTITUTE · Deepen

M32-IDN-003 — **I can use even/odd and cofunction identities to rewrite trigonometric expressions.**

SUPPORTING SKILLS

D	Test a formula or graph for even or odd symmetry. (first introduced in Math 31) SK-FUNC-SYMMETRY · Deepen
I	Apply sine/cosine cofunction identities. SK-TRIG-IDENTITY-COFUNCTION · Introduce
I	Apply even and odd trigonometric identities. SK-TRIG-IDENTITY-EVEN-ODD · Introduce
D	Substitute an identity to rewrite an expression. (first introduced in Math 32) SK-TRIG-IDENTITY-SUBSTITUTE · Deepen

M32-IDN-004 — **I can verify a trigonometric identity using the required identity families.**

SUPPORTING SKILLS

A	Simplify an algebraic expression using operation properties. (inherited from middle school) SK-ALG-EXPRESSION-SIMPLIFY · Apply
I	Choose a productive identity or algebraic manipulation. SK-TRIG-IDENTITY-STRATEGY · Introduce
D	Substitute an identity to rewrite an expression. (first introduced in Math 32) SK-TRIG-IDENTITY-SUBSTITUTE · Deepen
I	Maintain equivalence while transforming one side of an identity. SK-TRIG-IDENTITY-VERIFY · Introduce

SKILL PROGRESSION

I	Introduce
R	Reinforce
D	Deepen
A	Apply

Additional Trigonometric Identities

EXTENSION

LEARNING OBJECTIVES WITH SUPPORTING SKILLS

M32-IDX-001 I can apply double-angle identities.

SUPPORTING SKILLS

I	Apply double-angle identities. SK-TRIG-IDENTITY-DOUBLE-ANGLE · Introduce
A	Substitute an identity to rewrite an expression. (first introduced in Math 32) SK-TRIG-IDENTITY-SUBSTITUTE · Apply

M32-IDX-002 I can apply half-angle identities.

SUPPORTING SKILLS

A	Apply double-angle identities. (first introduced in Math 32) SK-TRIG-IDENTITY-DOUBLE-ANGLE · Apply
I	Apply half-angle identities. SK-TRIG-IDENTITY-HALF-ANGLE · Introduce

M32-IDX-003 I can apply angle sum and difference identities.

SUPPORTING SKILLS

A	Substitute an identity to rewrite an expression. (first introduced in Math 32) SK-TRIG-IDENTITY-SUBSTITUTE · Apply
I	Apply angle sum and difference identities. SK-TRIG-IDENTITY-SUM-DIFFERENCE · Introduce

M32-IDX-004 I can determine exact trigonometric values using extension identities.

SUPPORTING SKILLS

D	Express common trigonometric values exactly. (first introduced in Math 32) SK-TRIG-EXACT-VALUE · Deepen
A	Apply double-angle identities. (first introduced in Math 32) SK-TRIG-IDENTITY-DOUBLE-ANGLE · Apply
A	Apply half-angle identities. (first introduced in Math 32) SK-TRIG-IDENTITY-HALF-ANGLE · Apply
A	Apply angle sum and difference identities. (first introduced in Math 32) SK-TRIG-IDENTITY-SUM-DIFFERENCE · Apply

SKILL PROGRESSION

I	Introduce
R	Reinforce
D	Deepen
A	Apply

Triangle Area from SAS

EXTENSION

LEARNING OBJECTIVES WITH SUPPORTING SKILLS

M32-TRX-001	**I can calculate the area of a triangle from two sides and the included angle.**
SUPPORTING SKILLS	
A	Express common trigonometric values exactly. (first introduced in Math 32) SK-TRIG-EXACT-VALUE · Apply
I	Calculate triangle area from two sides and the included angle. SK-TRIG-TRIANGLE-AREA-SAS · Introduce
I	Classify triangle data as SSS, SAS, ASA, AAS, or SSA. SK-TRIG-TRIANGLE-DATA · Introduce

SKILL PROGRESSION

I Introduce · R Reinforce · D Deepen · A Apply

Polar Coordinates

REQUIRED

LEARNING OBJECTIVES WITH SUPPORTING SKILLS

M32-POL-001 — **I can plot and identify points in polar coordinates.**

SUPPORTING SKILLS

I	Plot a point from radius and angle. SK-POL-PLOT · Introduce
A	Determine a reference angle and its quadrant. (first introduced in Math 32) SK-TRIG-ANGLE-REFERENCE · Apply

M32-POL-002 — **I can represent a polar point using equivalent coordinate pairs.**

SUPPORTING SKILLS

I	Generate equivalent polar coordinate pairs. SK-POL-EQUIVALENT · Introduce
A	Generate and recognize coterminal angles. (first introduced in Math 32) SK-TRIG-ANGLE-COTERMINAL · Apply

M32-POL-003 — **I can convert points between polar and rectangular coordinates.**

SUPPORTING SKILLS

I	Use coordinate relationships to convert a point between systems. SK-POL-CONVERT-POINT · Introduce
A	Use coordinates and radius to evaluate trigonometric functions. (first introduced in Math 32) SK-TRIG-CIRCLE-POINT · Apply
A	Evaluate an inverse trigonometric expression exactly or approximately. (first introduced in Math 32) SK-TRIG-INVERSE-EVALUATE · Apply

Math 32 In-Depth Curriculum

Precalculus: Trigonometry

SKILL PROGRESSION

I Introduce · R Reinforce · D Deepen · A Apply

Polar Functions

EXTENSION

LEARNING OBJECTIVES WITH SUPPORTING SKILLS

M32-PLX-001 — **I can graph and interpret basic polar functions.**

SUPPORTING SKILLS

A	Identify input and output values in a formula, graph, table, or context. (first introduced in Math 12) SK-FUNC-INPUT-OUTPUT · Apply
D	Plot a point from radius and angle. (first introduced in Math 32) SK-POL-PLOT · Deepen
A	Locate quarter-period key points of a periodic function. (first introduced in Math 32) SK-TRIG-GRAPH-KEYPOINTS · Apply

M32-PLX-002 — **I can convert equations between polar and rectangular forms.**

SUPPORTING SKILLS

A	Simplify an algebraic expression using operation properties. (inherited from middle school) SK-ALG-EXPRESSION-SIMPLIFY · Apply
I	Substitute coordinate identities to convert an equation between systems. SK-POL-CONVERT-EQUATION · Introduce
A	Apply the Pythagorean trigonometric identities. (first introduced in Math 32) SK-TRIG-IDENTITY-PYTHAGOREAN · Apply

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Precalculus: Trigonometry

SKILL PROGRESSION

I Introduce · R Reinforce · D Deepen · A Apply

Vectors

EXTENSION

LEARNING OBJECTIVES WITH SUPPORTING SKILLS

M32-VEC-001 — I can represent a vector geometrically and in component form.

SUPPORTING SKILLS

A	Add congruence, parallel, perpendicular, and angle markings to a diagram. (first introduced in Math 22) SK-GEO-DIAGRAM-MARK · Apply
I	Write a vector in component form. SK-VEC-COMPONENTS · Introduce

M32-VEC-002 — I can add, subtract, and multiply vectors by scalars.

SUPPORTING SKILLS

A	Add, subtract, multiply, and divide signed integers. (inherited from middle school) SK-NUM-INTEGER-OPERATE · Apply
I	Perform component-wise vector operations. SK-VEC-OPERATE · Introduce

M32-VEC-003 — I can determine vector magnitude, direction, and unit vector.

SUPPORTING SKILLS

A	Calculate non-axis-aligned distance using the Pythagorean theorem or distance formula. (first introduced in Math 22) SK-COORD-DISTANCE-PYTHAGOREAN · Apply
I	Determine a vector's direction angle with correct quadrant. SK-VEC-DIRECTION · Introduce
I	Calculate vector magnitude using the distance formula. SK-VEC-MAGNITUDE · Introduce
I	Normalize a nonzero vector to obtain a unit vector. SK-VEC-UNIT · Introduce

M32-VEC-004 — I can resolve a vector into horizontal and vertical components.

SUPPORTING SKILLS

A	Match opposite, adjacent, and hypotenuse sides to a trigonometric ratio. (first introduced in Math 22) SK-TRI-TRIG-RATIO · Apply
D	Write a vector in component form. (first introduced in Math 32) SK-VEC-COMPONENTS · Deepen
A	Determine a vector's direction angle with correct quadrant. (first introduced in Math 32) SK-VEC-DIRECTION · Apply

M32-VEC-005 — I can perform basic operations with vectors in three dimensions.

SUPPORTING SKILLS

I	Represent and operate on vectors with three components. SK-VEC-3D · Introduce
D	Calculate vector magnitude using the distance formula. (first introduced in Math 32) SK-VEC-MAGNITUDE · Deepen
D	Perform component-wise vector operations. (first introduced in Math 32) SK-VEC-OPERATE · Deepen

Math 32 In-Depth Curriculum

Precalculus: Trigonometry

SKILL PROGRESSION

I	Introduce
R	Reinforce
D	Deepen
A	Apply

Parametric Equations

EXTENSION

LEARNING OBJECTIVES WITH SUPPORTING SKILLS

M32-PAR-001 **I can graph a plane curve defined by parametric equations with its orientation.**

SUPPORTING SKILLS

A	Identify input and output values in a formula, graph, table, or context. (first introduced in Math 12) SK-FUNC-INPUT-OUTPUT · Apply
I	Graph a parametric curve with its direction of motion. SK-PAR-GRAPH · Introduce
I	Generate ordered pairs from parametric equations. SK-PAR-TABLE · Introduce

M32-PAR-002 **I can eliminate a parameter to obtain a rectangular equation.**

SUPPORTING SKILLS

A	Isolate a specified variable in an equation or formula. (first introduced in Math 12) SK-ALG-VARIABLE-ISOLATE · Apply
I	Eliminate a parameter algebraically. SK-PAR-ELIMINATE · Introduce
A	Substitute an equivalent expression for a variable. (first introduced in Math 12) SK-SYS-SUBSTITUTE · Apply

M32-PAR-003 **I can create or interpret parametric equations for motion in the plane.**

SUPPORTING SKILLS

A	Explain a model parameter in the language of its context. (first introduced in Math 21) SK-MODEL-PARAMETER-INTERPRET · Apply
I	Interpret or construct parametric equations for planar motion. SK-PAR-MODEL · Introduce
D	Generate ordered pairs from parametric equations. (first introduced in Math 32) SK-PAR-TABLE · Deepen
A	Attach and interpret units for rates, inputs, and outputs. (first introduced in Math 12) SK-REP-UNITS · Apply